🧪 Atomic Structure

1. Introduction

Atomic structure is the foundation of chemistry and physics. It explains how matter is built, why elements behave differently, and how chemical reactions occur. Every atom consists of a dense nucleus surrounded by electrons in defined energy levels.

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2. Components of the Atom

- **Protons (p^+)**: Positively charged particles in the nucleus; determine the element's identity.
- **Neutrons (n^0)**: Neutral particles in the nucleus; add mass and stabilize the atom.
- **Electrons (e^-)**: Negatively charged particles moving in orbitals around the nucleus; responsible for bonding and reactivity.
- **Nucleus**: Contains nearly all the atom's mass, yet is extremely small compared to the atom's overall size.

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3. Atomic Number and Mass

- **Atomic Number (Z)** = number of protons.
- **Mass Number (A)** = protons + neutrons.
- **Isotopes**: Atoms of the same element with different neutron counts.
- **Relative Atomic Mass**: Weighted average of isotopes found in nature.

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4. Electron Arrangement

- Electrons occupy **energy levels (shells)** around the nucleus.
- Each shell has a maximum capacity (2, 8, 18, ...).
- **Valence electrons** (outermost shell) determine chemical bonding and reactivity.
- The arrangement follows the **Aufbau principle, Pauli exclusion principle, and Hund's rule**.

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5. Models of Atomic Structure

- **Dalton's Model**: Atoms are indivisible particles.
- **Thomson's Model**: "Plum pudding" model with electrons embedded in positive charge.
- **Rutherford's Model**: Dense nucleus with electrons orbiting around it.
- **Bohr's Model**: Electrons in fixed energy levels.
- **Quantum Mechanical Model**: Modern view; electrons exist in orbitals defined by probability distributions.

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6. Importance of Atomic Structure

- Explains **periodic trends** (atomic size, ionization energy, electronegativity).
- Determines how atoms bond (ionic, covalent, metallic).
- Provides the basis for understanding **chemical reactions, materials science, and modern technology**.

📘 **Summary**

Atomic structure describes how protons, neutrons, and electrons are arranged in atoms. The nucleus holds mass and identity, while electrons define reactivity. From Dalton to quantum mechanics, models of the atom have evolved to explain matter's properties. Mastery of atomic structure is essential for chemistry, physics, and all sciences dealing with matter.